

GAWTAM CHITHRA RAMESH

+46-76541 8589 gawtamcr3@gmail.com gawtamcr gawtamcr

Education

KTH Royal Institute of Technology

Aug. 2024 – Jul. 2026

Master of Science in [Systems, Controls, and Robotics](#)

Stockholm, Sweden

- **Coursework:** Robot Learning & Embodied AI, Deep Learning Advanced, Image Analysis & Computer Vision, Introduction to Robotics, Multi-Agent Systems.

Indian Institute of Technology Madras

Aug. 2019 – Jul. 2024

Dual Degree (B.Tech in [Engineering Design](#) and M.Tech in [Robotics](#))

Chennai, India

- **Coursework:** Mechanics of Serial Robots, Field and Service Robotics, Control Systems, Fundamentals of Deep Learning, Data Structures and Algorithms.

Professional Experience

Master's Thesis

Jan 2026 – Jun 2026

ABB Robotics, Mentors: [Matthew Lock](#) and [Jonathan Styrd](#)

Västerås, Sweden

- Architecting a System 1 / System 2 framework for industrial manipulation: a generative flow-matching trajectory planner (System 1) steered at inference time by Signal Temporal Logic robustness gradients (System 2), replacing black-box policies with a verifiable, explainable intermediate layer.
- Designed a single generative model that can be steered by arbitrary STL specifications at inference time, eliminating retraining or per-task expert demonstrations and enabling flexible transfer across task variants.
- Validated the framework with full specification satisfaction on reach-avoid benchmarks; benchmarking head-to-head against state-of-the-art STL planners (DAG-STL, ZSTP) on Maze2D and AntMaze, targeting a top-venue robotics submission.
- Collaborating with ABB Robotics R&D on transfer of the approach to long-horizon 6-DoF manipulation use cases.

Device Research Intern, 3D Scene Graphs

Jul 2025 – Dec 2025

Ericsson Research, Mentors: [Püren Güler](#) and [Hector Caltenco](#)

Lund, Sweden

- Developed a robust 3D scene-graph pipeline by integrating instance segmentation and tracking into a SOTA framework (Hydra), reducing parameter sensitivity for reliable scene interpretation across diverse environments.
- Designed a synchronized RGB-D inpainting pipeline to remove privacy-sensitive objects from 3D scene graphs in real time, addressing trustworthy-AI requirements for human-centered deployment, and benchmarked SOTA inpainting algorithms balancing reconstruction fidelity with real-time performance.
- Lead author on submissions to **ICPR 2026** (under review) and **IMPROVE 2026** (accepted).

Graduate Robotics Intern, Modular Collaborative Robots

Dec 2022 – Jul 2023

Systemantics India Pvt. Ltd., Mentor: [Jagannath Raju](#)

Bangalore, India

- Developed backward-chained Behavior Trees on a 6-DoF industrial manipulator for robust, long-horizon trajectory planning in collaborative-robot deployments.
- Designed a 6-DoF trajectory generation and control pipeline in C++, achieving real-time joint actuation via low-latency D-Bus and SocketCAN communication with motor controllers.

Robotics Intern, Pallet Handling

Jun 2021 – Jul 2021

Yaskawa Pvt. Ltd., Mentor: Manju Tiwari

Gurgaon, India

- Programmed industrial 6-DoF manipulators for mixed-carton palletizing, a core machine-tending task, using MotoSim simulation and teach-pendant scripting.
- Built a C++ memory-optimization algorithm for streaming data between PC and robot controller, improving throughput in high-volume pick-and-place operations.

Research Experience

Graduate Research, Deformable Object Manipulation

Jan 2025 – Nov 2025

[Robotics, Perception and Learning \(RPL\)](#), KTH, Advisor: [Florian T. Pokorny](#)

Stockholm, Sweden

- Integrated taxonomy-guided vision-language models (Gemini, Qwen) to interpret deformable-object manipulation scenes and map structured outputs to motion primitives.
- Built a prompt and taxonomy framework that generates robot-parsable, formally-structured specifications from natural-language prompts; iteratively improved the framework through quantitative failure-mode analysis and integration of multimodal and temporal cues.
- Lead author on the resulting submission to the ROMADO workshop at **IROS 2025**.

Dual Degree Project, Structure from Motion for Autonomous Ultrasound

Jan 2024 – May 2024

INSPIRE Lab, Advisor: Nirav Patel

IIT Madras, India

- Built a 3D perception system using Structure from Motion (COLMAP) on monocular RGB images for cost-efficient human torso reconstruction, enabling more accessible ultrasound diagnostics.
- Designed an autonomous scanning system integrating a 5-DoF robotic arm with single-camera RGB perception via ROS, MoveIt, and Gazebo simulation.
- Implemented an end-to-end perception-to-planning loop: 3D point-cloud generation, trajectory planning, and ultrasound-probe execution.

Projects

Robot Learning and Embodied AI | DD2601, KTH

Sep 2025 – Oct 2025

- Built a 3D open-vocabulary semantic mapping pipeline using RGB-D data from ARKitScenes, integrating OwlV2 detection, SAM segmentation, and CLIP grounding with depth-fusion 2D-to-3D projection for queryable scene maps.
- Deployed a pre-trained Vision-Language-Action (VLA) model for robot manipulation, executing policy inference from visual inputs and natural-language commands; analyzed generalization and robustness limits on unseen tasks.

Reinforcement Learning for Pacman Capture-the-Flag | KTH

Mar 2025 – May 2025

- Designed a hybrid agent for Pacman Capture-the-Flag in Unity3D, merging a custom Behavior Tree for high-level strategy with role-specific PPO policies for specialized actions (attack, defend, escape) in a partially-observable, dynamic, multi-agent environment.
- Trained the RL policies via Unity ML-Agents using phased training, self-play, and curriculum learning, eliciting adaptive behaviour and on-the-fly role switching across competitive matches.

Introduction to Robotics | DD2410, KTH

Sep 2024 – Oct 2024

- Implemented an iterative inverse-kinematics solver from scratch for a 7-DoF KUKA arm in ROS.
- Integrated a Behavior Tree mission planner for a TIAGo mobile manipulator in ROS/Gazebo, enabling autonomous navigation, vision-based grasping, and recovery from kidnapped-robot scenarios.
- Engineered 2D occupancy-grid mapping in ROS from laser scans; applied RRT* for non-holonomic Dubins-car path planning.

Multi-Agent RL for VR Shooter | CFI, IIT Madras

May 2020 – Aug 2020

- Built multi-agent and cooperative-play PPO policies in Unity3D for a shooter strategy game with obstacles and procedurally generated mazes, achieving superhuman gameplay against human opponents.
- Integrated the trained agents into a Unity VR environment for direct virtual-reality interaction with the learned policies.

Achievements

- Awarded the **IIT Madras Young Research Fellowship** (top 30 of 800+ applicants) for outstanding research potential.
- Top-percentile performance in India's national entrance examinations: **97.64 percentile in JEE Advanced** and **99.10 percentile in JEE Mains** (out of 1.5 million applicants); **All-India Rank 52 of 50,000+** in UCEED (Design).

Leadership & Mentoring

Head of Industrial & Public Relations | Department of Engineering Design, IIT Madras

Apr 2021 – May 2022

- Led a three-tier team running placement and internship processes for 1,200+ students; designed and executed the department's first hybrid recruitment framework, expanding alumni and industry outreach to broaden the opportunity pipeline.

Strategist | Computer Vision & Intelligence Group, IIT Madras

Apr 2020 – May 2021

- Co-organized a Deep Learning summer school for 500+ students covering DL, RL, and computer vision; co-mentored student projects on image analysis and reinforcement learning for games.

Graduate Teaching Assistant | Field and Service Robotics (ID4060), IIT Madras

Aug 2023 – Nov 2023

- Designed and graded ROS and CoppeliaSim assignments for 40 students; ran small-group reviews on assignments and end-of-term projects across diverse student backgrounds.

Technical Skills

- **Languages:** Python, C++, C, C# **Machine Learning:** PyTorch, JAX, OpenCV, wandb
- **Robotics & Simulation:** ROS / ROS2, MoveIt, Gazebo, MuJoCo, PyBullet, Unity3D, Unity ML-Agents, CoppeliaSim, MATLAB
- **Tools:** Docker, Git, L^AT_EX, CUDA, Slurm, VS Code